

Advanced Database

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1. Disaster Recovery Plan (Financial Services Database)

RTO (Recovery Time Objective):

Maximum acceptable downtime is 1 hour. This ensures minimal disruption to financial services.

RPO (Recovery Point Objective):

Maximum acceptable data loss is 5 minutes. This ensures critical transaction data is preserved.

Backup Strategy:

- Full backups performed daily at midnight
- Incremental backups every 15 minutes
- Transaction log backups every 5 minutes
- Backups stored on-site and off-site (cloud)
- Backup data encrypted using AES-256
- Retention period: 30–90 days

Failover Procedures:

- Primary and secondary (replica) database setup
- Automatic failover enabled
- Real-time data replication
- Load balancer redirects traffic to standby system during failure

Communication Plan:

- Incident response team includes IT, security, and management
- Immediate alerts sent upon failure detection
- Stakeholders updated every 30 minutes
- Communication via email, SMS, and dashboards
- Post-incident report generated within 24 hours

Testing and Validation:

- Disaster recovery drills conducted quarterly
- Backup restoration tested monthly
- Failover simulations performed twice a year
- Ensure RTO and RPO targets are met

2. Security Audit Implementation

Role-Based Access Control:

- Roles defined: Admin, Analyst, User
- Least privilege principle applied
- Multi-factor authentication enforced

Data Encryption:

- Data encrypted at rest using AES-256
- Data encrypted in transit using TLS/SSL
- Sensitive data (passwords, account numbers) secured

Audit Logging:

- Logs include login attempts, data access, and changes
- Logs stored securely and retained for at least 1 year
- Real-time alerts for suspicious activity

SQL Injection Prevention:

- Use of parameterized queries
- Input validation and sanitization
- Avoid dynamic SQL
- Use ORM frameworks where applicable

Compliance Reporting:

- Compliance with PCI-DSS and GDPR standards
- Monthly security and audit reports generated
- Includes access logs, incidents, and risk assessments